

DATASET REPORT

Algeria Wildfire Risk Dataset

Structure, labels and predictors of the cell-day table

OBSERVATION UNIT	Cell-day (grid cell x date)
SPATIAL RESOLUTION	Approx. 0.1° x 0.1° grid cells
TARGET	label (detection-based, binary)
COVERAGE	Algeria

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1. Dataset Overview

What the dataset contains, what one row represents, and what is documented in this report.

The Algeria Wildfire Risk dataset is a table of **cell-days**. Each row joins one geographical grid cell of approximately $0.1^\circ \times 0.1^\circ$ to one particular date, and records the conditions associated with that place on that day together with a fire label.

Each row combines several layers of information: environmental measurements, vegetation state, historical fire context and a detection-based label. The dataset is therefore both a map and a calendar: the same cell appears on many dates, and many cells appear on the same date.

Dataset at a glance

Item	Description
Observation unit	One grid cell on one date (a cell-day)
Spatial resolution	Approximately $0.1^\circ \times 0.1^\circ$ cells across Algeria
Identifiers	cell_id identifies the cell; lat and lon give the cell centre
Target	label: whether fire detection is recorded for the cell-day
Predictor families	Meteorology, recent weather history, Fire Weather Index (FWI) variables, vegetation, historical fire information
Removed columns	sample_kind, n_detections, max_frp (target leakage)
Structure	Panel-like: many places observed over many dates

Reading guide

Sections 2 to 4 describe the structure, the predictor families and the meaning of the label. Section 5 covers the FWI variables, Section 6 covers weather history, Section 7 documents the leakage audit, and Sections 8 and 9 give modelling considerations and a column contract template.

2. Data Structure: The Cell-Day

How a single row is assembled and why its meaning depends on the observation date.

Identifier and location columns

Column	Role
cell_id	Identifies the spatial cell. The same value repeats across dates.
lat, lon	Coordinates of the cell centre. They support mapping and spatial context and do not describe what happened on a given day.
label	Detection-based target for the cell-day (see Section 4).

Key structural properties

- **Repeated cells.** A repeated cell_id is the record of a changing place, not duplication. Weather, vegetation and history change from date to date.
- **Shared dates.** Different cells appear on the same date, and nearby cells can share weather systems, vegetation patterns and fire conditions.
- **Non-independence.** Rows of the same cell are connected, and rows of neighbouring cells may be connected. The data has a panel-like structure across space and time.
- **Date dependence.** A row cannot be interpreted without its date. The target can change from day to day and is not a permanent property of a cell.
- **Approximate location.** A 0.1° cell is a practical spatial unit. It does not claim that every location inside the cell behaves identically.

Information horizon

Every row is a bundle of measurements with an information horizon. The relevant question for each value is not only what it describes, but when it could have been known.

3. Feature Families

The kinds of information carried by each row and their relationship to the prediction moment.

Family	What it describes	Timing status
Spatial	Where the observation is located (cell_id, lat, lon).	Static context
Meteorology	Weather conditions recorded for the observation date.	Valid if available at prediction time
Weather history	Conditions accumulated over preceding days, such as persistent dryness.	Valid only if built from prior observations
Fire Weather Index	Structured indices of fire-weather danger (ffmc, dmc, dc, isi, bui, fwi).	Valid if timing is properly defined
Vegetation	The state of the fuel environment.	Valid if timing is properly defined
Historical fire	Earlier fire events, e.g. days_since_last_fire.	Valid only if computed strictly from prior observations
Same-day detection	Satellite fire detection on the target day (sample_kind, n_detections, max_frp).	Removed: reveals the target

These families are not interchangeable. Spatial information identifies the setting, temporal information places the observation in the calendar, meteorological variables describe current conditions, weather history carries accumulated information, vegetation describes the fuel, and historical fire information provides earlier context.

The cell-day structure lets these layers meet without collapsing them into a single idea of risk. A dry period may affect several neighbouring cells without affecting them identically, vegetation may differ between cells, and a previous fire may change the history of one cell and leave another untouched.

4. The Target Label

The label records satellite detection, which is not the same as physical fire occurrence.

Definition

Label is a **detection-based target**. It indicates whether the dataset contains evidence of wildfire detection for a given cell and date. It is a statement produced by an observation process, not a complete account of everything that physically happened in the cell.

Value	Meaning	Does not mean
label = 1	Fire activity was detected for this cell-day.	Fire size, duration, intensity or total damage.
label = 0	The dataset has no qualifying detection for this cell-day.	That no fire physically existed in the cell.

From fire to label

A physical fire becomes a label only if the whole chain holds: combustion occurs in the cell, its signature is strong and exposed enough to be registered by a satellite-based system at the right time, and the detection is assigned to the cell and date used by the dataset. The chain can break at several points:

- a small fire may stay below the detection threshold;
- cloud or smoke may interfere with what the sensor sees;
- a fire may start and end between observations;
- a fire near the edge of a cell may be assigned less exactly.

Interpretation

A prediction for label = 1 concerns the likelihood of a recorded detection under the dataset's observation system. A prediction for label = 0 concerns the absence of recorded detection, not proof of an untouched landscape.

5. Fire Weather Index Variables

Six related variables that describe different parts of the fuel-and-weather system.

The FWI variables form a chain from fine-fuel moisture to an overall fire-weather measure. They are not interchangeable versions of one measurement, and `fwi` should be read together with its components.

Variable	Describes	Response time
<code>ffmc</code>	Moisture of fine fuels: small, easily ignited material.	Fast, follows recent weather
<code>dmc</code>	Moisture in deeper organic layers.	Intermediate
<code>dc</code>	Longer-term drying in heavier materials.	Slow, retains memory of prolonged drying
<code>isi</code>	Fine-fuel moisture combined with wind-related fire spread potential.	Spread potential
<code>bui</code>	Amount of available fuel, from the deeper moisture components.	Fuel availability
<code>fwi</code>	Overall fire-weather measure combining spread and fuel availability.	Summary index

Interpretation notes

- **FWI is not an ignition probability.** It describes weather-related fire danger, not the full process that produces a labelled cell-day.
- **Similar `fwi`, different conditions.** Two cell-days can share a similar `fwi` while differing in fine-fuel dryness, deeper-layer dryness, spread potential and available fuel.
- **Complementary families.** FWI describes weather and drying, vegetation describes available fuel, and fire history supplies temporal context. Each answers a different question.

6. Weather History and Temporal Validity

How recent weather is represented without importing information from the future.

Fuels respond more slowly than the atmosphere. Wind can change within hours, while vegetation moisture and surface dryness reflect a longer chain of conditions. A same-day temperature value can therefore describe different situations depending on what came before it, and weather history carries information that same-day weather cannot.

Historical features are valid only if their contents were available at prediction time. The following checklist applies to every weather-history value:

Check	Question
1. Prediction date	Which date is being predicted?
2. Prior data only	Was the value formed only from observations before that date?
3. Clear window	Is the time window of the calculation clearly defined?
4. No target day	Does the calculation include the target day or later records?

Typical failure cases

- **Hidden inclusion.** A feature described as recent whose construction includes the observation date, or data measured after the event.
- **Future contamination.** `days_since_last_fire` computed over the full dataset can find a later event. It is valid only when calculated strictly from prior observations.
- **Same value, different history.** Two rows with the same current measurement can carry different environmental histories.

Principle

History is informative because it leaves traces in the present. Future information is dangerous because it leaves traces that would not yet have been visible.

7. Target Leakage Audit

Three original columns were removed from the working dataset because they reveal the target.

Same-day detection information describes what happened on the very day represented by the target. Used as a predictor, it turns prediction into recognition: the model reads evidence that the event has already been detected.

Column	Content	Decision
<code>sample_kind</code>	Type of record, tied to fire-detection sampling and to how the target was constructed.	Removed (leakage)
<code>n_detections</code>	Number of same-day satellite fire detections for the cell-day.	Removed (leakage)
<code>max_frp</code>	Maximum fire radiative power of the same-day detections.	Removed (leakage)

Why it matters

- **Distorted feature importance.** Detection fields can eclipse meteorology, weather history, FWI, vegetation and history.
- **Inflated scores.** With random splits, the same cells and nearby dates appear in training and test data, so the score reflects recognition of recorded fire rather than a prospective risk signal.
- **Misleading explanations.** `max_frp` reports the intensity of detected activity. It does not explain why vegetation became combustible.
- **Metadata can leak.** A field that only describes how a row was sampled, such as `sample_kind`, can reveal the target as much as a physical measurement.

What remains valid

Removal does not make the columns meaningless: their relationship with the target is exactly why they are dangerous. They remain documented for the original dataset. `days_since_last_fire` is treated differently because it is historical information and is valid when computed strictly from prior observations. The boundary between a useful and a forbidden feature is the column's position in time, not its subject.

8. Modelling Considerations

Practical implications of the dataset design for anyone building models on it.

Topic	Consideration
Validation	Cells and dates are not independent. Random splits can place the same cells and nearby dates in both training and test data and inflate performance.
Target meaning	Models learn patterns associated with recorded detection, not a catalogue of every fire in Algeria.
Temporal features	Every history feature must be built from observations before the prediction date.
Coordinates	lat and lon locate observations. They do not replace environmental and historical features.
FWI reading	Use the components (ffmc, dmc, dc, isi, bui) alongside fwi rather than as a single danger score.
Regional differences	Northern Mediterranean areas and the southern regions differ substantially, so a familiar index may carry a different meaning from one cell to the next.
Observation limits	Cloud, smoke, sensor thresholds and timing can change what gets labelled, which matters when comparing regions and dates.

9. Column Contract

A template to document, for every label and feature, its source and the time window it is allowed to see.

For each column, record its data source and permitted time window, then run a sanity check such as plotting the target against the most suspicious features to confirm the feature reflects the fire story it is assumed to.

Column	Source	Allowed time window	Status
label	Detection-based record	Target date	Target
days_since_last_fire	Historical fire records	Strictly before prediction date	Valid if prior only
ffmc, dmc, dc, isi, bui, fwi	FWI variables	Defined per variable	To be verified
sample_kind	Detection sampling	Same day as target	Removed
n_detections	Satellite detections	Same day as target	Removed
max_frp	Satellite detections	Same day as target	Removed